A hitch-hiker's guide to the metaplectic trace formula for $SL(2)$

Section 2, revised for the double cover (stage 1 draft)

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This is the revised Section 2 of the 1995 notes, restricted to $n = 2$, compiled on its own. Lemmas 2.1–2.9 keep the numbers of the 1995 notes; Lemma 2.10 is new. The changes are listed in `revision_log.md`.

2 Background on the metaplectic double cover

This section fixes the double cover of $G = \mathrm{Res}_{k/\mathbb{Q}} SL(2)$ and the conventions used in the rest of the paper. Section 2.1 treats a single place: Kubota's cocycle, the splittings over unipotent and compact subgroups, the center, and the preimages of tori. These are then assembled into the cover of $G(\mathbb{Q}_p)$. Section 2.2 builds the adelic cover and its splitting over $G(\mathbb{Q})$, and Section 2.3 fixes a double cover of $GL(2)$ and the resulting $GL(2)$-action on the cover of $SL(2)$, which is what Section 7 uses.

From now on $n = 2$. The 1995 notes treated the n-fold cover; for $n = 2$ several statements become simpler, and one, the commutativity of preimages of abelian subgroups (Lemma 2.2(g)), becomes a two-line argument. What changes for general n is listed in Remark 2026.4.

Notation

Let k be a number field. (For $n = 2$ the hypothesis $\mu_n \subset k$ of the 1995 notes always holds, and k may have real places.) Let Σ_k be the set of places of k, Σ_k^∞ the archimedean places and Σ_k^f the finite places. Let $\mathbb{A}$ be the adele ring of $\mathbb{Q}$ and $\mathbb{A}_k$ that of k. For $v \in \Sigma_k$ let k_v be the completion. For $F = k_v$ non-archimedean let $\mathcal{O} = \mathcal{O}_F$ be its ring of integers, $\varpi = \varpi_F$ a uniformizer, q the order of the residue field and $\mathrm{ord} = \mathrm{ord}_F \colon F^\times \to \mathbb{Z}$ the normalized valuation, so $\mathrm{ord}(\varpi) = 1$ and $|\varpi| = q^{-1}$. We call v *dyadic* if $v \mid 2$, and put $e_v = \mathrm{ord}_v(2)$. Absolute values are normalized; at a complex place $|z|_v = z\bar{z}$.

Let $(a, b) = (a, b)_v$ be the quadratic Hilbert symbol of $F = k_v$ [W1, Se]. We use only the following properties:

(H1) $(\cdot, \cdot)$ is bimultiplicative and symmetric, with values ± 1; hence $(a^{-1}, b) = (a, b)$.

(H2) $(a, b) = 1$ if a or b lies in $F^{\times 2}$; $(a, -a) = 1$; and $(a, 1 - a) = 1$ for $a \neq 0, 1$ (the Steinberg relation).

(H3) If v is finite and not dyadic and $a, b \in \mathcal{O}^\times$, then $(a, b) = 1$. If v is complex, $(\cdot, \cdot) \equiv 1$. If v is real, $(a, b) = -1$ exactly when $a < 0$ and $b < 0$.

(H4) (Product formula) For $a, b \in k^\times$, $(a, b)_v = 1$ for almost all v and $\prod_v (a, b)_v = 1$.

By (H3), $(-1,-1)_v = 1$ at non-dyadic finite places and at complex places, and $(-1,-1)_v = -1$ at real places. At a dyadic place both values occur; $(-1,-1)_{\mathbb{Q}_2} = -1$.

Let G be the group over $\mathbb{Q}$ obtained from $SL(2)$ by restriction of scalars from k to $\mathbb{Q}$, so $G(\mathbb{Q}) = SL(2,k)$ and $G(\mathbb{Q}_p) = \prod_{v|p} SL(2,k_v)$.

2.1 Local aspects

Here $F = k_v$ for one place v. We define the cover, compute it on the subgroups that matter later (unipotent, compact, diagonal, central, anisotropic), and then pass to $G(\mathbb{Q}_p)$.

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,F)$ put $x(g) = c$ if $c \neq 0$ and $x(g) = d$ if $c = 0$, and define $\beta = \beta_v \colon SL(2,F) \times SL(2,F) \to \mu_2$ by

$$\beta(g_1,g_2) = \Big(\frac{x(g_1g_2)}{x(g_1)}, \frac{x(g_1g_2)}{x(g_2)}\Big) = \big(x(g_1), x(g_2)\big)\big(-x(g_1)^{-1}x(g_2),\, x(g_1g_2)\big). \tag{2.1}$$

At a complex place $\beta \equiv 1$ by (H3).

Lemma 2.1 (Kubota [K1], Moore [M]). *β is a continuous Borel 2-cocycle, μ_2 carrying the discrete topology. Its class in $H^2(SL(2,F),\mu_2)$ defines a topological central extension*

$$1 \to \mu_2 \to \overline{SL(2,F)} \xrightarrow{p} SL(2,F) \to 1,$$

which is nontrivial unless $F = \mathbb{C}$.

The two expressions in (2.1) agree, and the cocycle identity holds; both are checked on random elements over $\mathbb{R}$, $\mathbb{Q}_2$, $\mathbb{Q}_3$, $\mathbb{Q}_5$ and $\mathbb{Q}_7$ in `checks/sec2_n2_check.py`. For $F = \mathbb{R}$ the group $\overline{SL(2,\mathbb{R})}$ is the metaplectic group $Mp(2,\mathbb{R})$.

For a subgroup $H \subset SL(2,F)$ put $\overline{H} = p^{-1}(H)$. Elements of $\overline{SL(2,F)}$ are written (g,ζ) with $g \in SL(2,F)$, $\zeta \in \mu_2$, and multiplied by

$$(g_1,\zeta_1)(g_2,\zeta_2) = (g_1g_2,\ \beta(g_1,g_2)\zeta_1\zeta_2). \tag{2.2}$$

Since $\zeta^{-1} = \zeta$, inverses and conjugates are

$$(g,\zeta)^{-1} = (g^{-1}, \beta(g,g^{-1})\zeta), \qquad (g,\zeta)(h,\zeta')(g,\zeta)^{-1} = \big(ghg^{-1},\ \beta(g,h)\beta(gh,g^{-1})\beta(g,g^{-1})\zeta'\big). \tag{2.3}$$

Let $B \subset SL(2)$ be the upper triangular Borel subgroup, N its unipotent radical and A the diagonal torus. Write $a(t) = \mathrm{diag}(t,t^{-1})$ and $n(u) = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}$. For $\theta \in F^\times \smallsetminus F^{\times 2}$ let

$$T_\theta = \left\{ \begin{pmatrix} a & b\theta \\ b & a \end{pmatrix} \;\middle|\; a,b \in F,\ a^2 - \theta b^2 = 1 \right\} \cong E^1, \qquad E = F(\sqrt{\theta}),$$

the norm-one group of E. Every anisotropic maximal torus of $SL(2,F)$ is $GL(2,F)$-conjugate to some T_θ.

If v is finite let $K = K_0 = SL(2,\mathcal{O})$ and, for $m \geq 1$,

$$K_m = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathcal{O}) \;\middle|\; a-1,\ d-1,\ b,\ c \in \varpi^m\mathcal{O} \right\}.$$

If v is non-dyadic, define $\kappa = \kappa_v \colon K \to \mu_2$ by

$$\kappa\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} (c,d)_v, & 0 < |c|_v < 1, \\ 1, & \text{otherwise.} \end{cases} \tag{2.4}$$

If v is archimedean let $K = SO(2)$ (v real) or $K = SU(2)$ (v complex).

Lemma 2.2. *Let $F = k_v$.*

(a) $\beta \equiv 1$ on $N(F) \times N(F)$, so $n \mapsto (n,1)$ is a homomorphism and $\overline{N(F)} = N(F) \times \mu_2$.

(b) Let v be finite and non-dyadic. Then $\beta(k_1,k_2) = \kappa(k_1)\kappa(k_2)\kappa(k_1k_2)$ for $k_1, k_2 \in K$, so $k \mapsto (k, \kappa(k))$ is a homomorphism and $\overline{K} = K^* \times \mu_2$ with $K^* = \{(k,\kappa(k))\}$. The restriction of β to $K \times K$ is not identically 1, so the set $\{(k,1) \mid k \in K\}$ is not a subgroup.

(c) Let v be finite and $m \geq 1$ with $1 + \varpi^m \mathcal{O} \subset F^{\times 2}$. Then $\beta \equiv 1$ on $K_m \times K_m$, so $k \mapsto (k,1)$ splits $\overline{K_m}$. The hypothesis holds for every $m \geq 1$ if v is non-dyadic, and for $m \geq 2e_v + 1$ if v is dyadic.

(d) Let v be archimedean. If v is complex, $\overline{SL(2,F)} = SL(2,F) \times \mu_2$. If v is real, $\beta \equiv 1$ on $A(\mathbb{R})^0 \times A(\mathbb{R})^0$, so $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$, while $\overline{SO(2)}$ is connected, the nontrivial double cover of the circle [Ge2].

(e) The center of $\overline{SL(2,F)}$ is $\overline{\{\pm 1\}}$, a group of order 4 in which $(-1,\zeta)^2 = (1,(-1,-1)_v)$. It is cyclic when $(-1,-1)_v = -1$ (for example $F = \mathbb{R}$ or $F = \mathbb{Q}_2$) and is $\{\pm 1\} \times \mu_2$ otherwise.

(f) On $A(F)$, $\beta(a(t_1), a(t_2)) = (t_1,t_2)_v$. Hence $\overline{A(F)}$ is abelian, and $(a(t),\zeta)^2 = (a(t^2),(t,-1)_v)$.

(g) If $g_1, g_2 \in SL(2,F)$ commute, then $\beta(g_1,g_2) = \beta(g_2,g_1)$. Consequently $\overline{H}$ is abelian for every abelian subgroup $H \subset SL(2,F)$; in particular $\overline{A(F)}$ and every $\overline{T_\theta}$ are abelian.

(h) Let $U = gNg^{-1}$ with $g \in SL(2,F)$. Then $u \mapsto (u,1)$ is the unique homomorphic section of $\overline{U} \to U$. Consequently, for every $(g,\zeta) \in \overline{SL(2,F)}$ and $u \in N(F)$,

$$(g,\zeta)(u,1)(g,\zeta)^{-1} = (gug^{-1},1).$$

Proof. (a) For $n_1, n_2 \in N(F)$ all three values of x equal 1.

(b) This is Kubota's computation [K2]; see also [Ge2]. The identity is also checked numerically for $F = \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7$, together with the existence of pairs in $K \times K$ with $\beta = -1$.

(c) Let $k_i = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \in K_m$ for $i = 1,2$, and $k_1k_2 = \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix}$. All a_i and d_i lie in $1 + \varpi^m\mathcal{O} \subset F^{\times 2}$, and $c_3 = c_1a_2 + d_1c_2$. If $c_1, c_2, c_3 \neq 0$, then $c_1a_2/c_3 + d_1c_2/c_3 = 1$, so by (H2) and (H1)

$$\beta(k_1,k_2) = \left(\frac{c_3}{c_1}, \frac{c_3}{c_2}\right) = \left(\frac{c_1a_2}{c_3}, \frac{d_1c_2}{c_3}\right) = 1,$$

because a_2 and d_1 are squares. If $c_3 = 0 \neq c_1c_2$, then $x(k_1k_2) = d_3$ is a square and $c_2 = -c_1a_2/d_1$, so $\beta = (c_1,c_2) = (c_1,-c_1)(c_1,a_2/d_1) = 1$. If $c_1 = 0 \neq c_2$, then $x(k_1) = d_1$ is a square and $c_3 = d_1c_2$, so $\beta = (c_3/d_1,\, d_1) = 1$; the case $c_2 = 0 \neq c_1$ is the same with a_2 in place of d_1. If

$c_1 = c_2 = 0$, all x-values are squares. Finally, $1+\varpi\mathcal{O} \subset F^{\times 2}$ for non-dyadic F by Hensel's lemma, and $1+4\varpi\mathcal{O} \subset F^{\times 2}$ for dyadic F by the binomial series for $(1+4\varpi y)^{1/2}$; and $4\varpi\mathcal{O} = \varpi^{2e_v+1}\mathcal{O}$.

(d) At a complex place $\beta \equiv 1$ by (H3). For $t_1, t_2 > 0$, $(t_1, t_2)_{\mathbb{R}} = 1$ by (H3) and (f). The statement about $SO(2)$ is standard [Ge2].

(e) Commuting elements give commuting lifts by (g), so $\overline{\{\pm 1\}}$ is central; conversely, the image of a central element is central in $SL(2,F)$. Since $x(-1) = -1$ and $x(1) = 1$, $(-1,\zeta)^2 = (1, \beta(-1,-1)) = (1, (-1,-1)_v)$ by (2.1). A group of order 4 with an element of order 4 is cyclic, and one of exponent 2 is $\{\pm 1\} \times \mu_2$.

(f) Here $x(a(t)) = t^{-1}$, so (2.1) gives $\beta(a(t_1), a(t_2)) = (t_2^{-1}, t_1^{-1}) = (t_1, t_2)$ by (H1). The square is $(a(t^2), (t,t))$, and $(t,t) = (t,-1)$ by (H2).

(g) If $g_1 g_2 = g_2 g_1$, then $x(g_1 g_2) = x(g_2 g_1)$, and the first expression in (2.1) is symmetric in g_1, g_2 because the quadratic Hilbert symbol is symmetric (H1). By (2.2), lifts (g_1, ζ_1) and (g_2, ζ_2) commute if and only if $\beta(g_1, g_2) = \beta(g_2, g_1)$.

(h) If $g \in B(F)$, then $U = N$ and the claim is (a). Otherwise g has lower left entry $\gamma \neq 0$, and $u_s = g\, n(s)\, g^{-1}$ has lower left entry $-\gamma^2 s$. So for $s_1, s_2 \neq 0$ with $s_1 + s_2 \neq 0$,

$$\beta(u_{s_1}, u_{s_2}) = \Big(\frac{s_1+s_2}{s_1}, \frac{s_1+s_2}{s_2}\Big) = \Big(\frac{s_1}{s_1+s_2}, \frac{s_2}{s_1+s_2}\Big) = 1$$

by (H1) and the Steinberg relation (H2). If $s_1 + s_2 = 0$, then $x(u_{s_1} u_{s_2}) = 1$ and $\beta = (1/s_1, -1/s_1) = 1$. So $u \mapsto (u, 1)$ is a homomorphism on U. Two homomorphic sections differ by a continuous homomorphism $U \cong F \to \mu_2$, which is trivial because $F = 2F$. Conjugation by (g, ζ) carries the section $n \mapsto (n, 1)$ of $\overline{N}$ to a homomorphic section of $\overline{gNg^{-1}}$, which is therefore $u \mapsto (u, 1)$. □

Example 2.A. Take $F = \mathbb{R}$. By (e) and (f), the element $(a(-1), 1) = (-1, 1)$ has order 4, so $\overline{\{\pm 1\}} \cong \mathbb{Z}/4$ and $\overline{A(\mathbb{R})}$ is an abelian group which is not $A(\mathbb{R}) \times \mu_2$. By (d), $\overline{A(\mathbb{R})}$ is generated by $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$ and $(-1, 1)$. At a non-dyadic place, on the other hand, $(-1,-1)_v = 1$, so the center is $\{\pm 1\} \times \mu_2$; and $\overline{A(\mathcal{O}^\times)}$ is split by $a \mapsto (a, 1)$, because $(t_1, t_2)_v = 1$ for units by (H3). This is consistent with (b), since $\kappa \equiv 1$ on $A(\mathcal{O}^\times)$.

Remark 2026.1. For $F = \mathbb{Q}_2$, Lemma 2.2(c) gives the splitting $k \mapsto (k, 1)$ over K_3, and β is not identically 1 on $K_2 \times K_2$. The cover nevertheless splits over K_2, by $k \mapsto (k, (c,d)_2)$ for $c \neq 0$ and $k \mapsto (k, 1)$ for $c = 0$. This is the local form at 2 of the classical splitting over $\Gamma_1(4)$ [K2, Ge2]. We have verified it numerically only (`checks/sec2_n2_check.py`), and the same section fails on K_1. Nothing below depends on it: the rest of the paper uses only that some K_m splits at each dyadic place.

The cover of $G(\mathbb{Q}_p)$. Let $\overline{G(\mathbb{Q}_p)}$ be the extension of $G(\mathbb{Q}_p) = \prod_{v|p} SL(2, k_v)$ by μ_2 defined by the cocycle

$$\beta_p(g_1, g_2) = \prod_{v|p} \beta_v(g_{1,v}, g_{2,v}).$$

It is the *quotient* of $\prod_{v|p} \overline{SL(2,k_v)}$ by the kernel $\{(\zeta_v) \mid \prod_v \zeta_v = 1\}$ of the product map $\prod_{v|p} \mu_2 \to \mu_2$. (The 1995 notes call it a subgroup.) For $H \subset G(\mathbb{Q}_p)$ put $\overline{H} = p^{-1}(H)$. Let B, N, A be the $\mathbb{Q}$-subgroups of G obtained from those of $SL(2)$, and let $T = \prod_{v|p} T_v$ with each T_v a maximal torus of $SL(2, k_v)$. Let $K_p = \prod_{v|p} K_v$. For p odd, $K_p^* = \prod_{v|p} K_v^*$, with K_v^* as in Lemma 2.2(b), maps to a subgroup of $\overline{G(\mathbb{Q}_p)}$ which we also denote by K_p^*.

Lemma 2.3. *(a) $n \mapsto (n,1)$ splits $\overline{N(\mathbb{Q}_p)}$.*

(b) If p is odd, $\overline{K_p} = K_p^* \times \mu_2$.

(c) For every p, $k \mapsto (k,1)$ splits $\overline{K_{m,p}}$, where $K_{m,p} = \prod_{v|p} K_{m,v}$, as soon as $m \geq 2e_v + 1$ for every $v \mid p$; if p is odd, $m \geq 1$ suffices.

(d) If $p = \infty$, $\overline{A(\mathbb{R})^0} = A(\mathbb{R})^0 \times \mu_2$, where $A(\mathbb{R})^0$ is the identity component of $A(k \otimes \mathbb{R})$.

(e) The center of $\overline{G(\mathbb{Q}_p)}$ is $\overline{\{\pm 1\}^{S_p}}$, where $S_p = \{v \mid p\}$. It is abelian, and the square of a lift of $\varepsilon = (\varepsilon_v)$ is $\prod_{v|p,\ \varepsilon_v = -1} (-1,-1)_v$.

(f) $\beta_p(a_1, a_2) = \prod_{v|p} (t_{1,v}, t_{2,v})_v$ for $a_i = a(t_i) \in A(\mathbb{Q}_p)$, and $\overline{A(\mathbb{Q}_p)}$ is abelian.

(g) The preimage of any abelian subgroup of $G(\mathbb{Q}_p)$, in particular $\overline{T}$, is abelian, and Lemma 2.2(h) holds for $G(\mathbb{Q}_p)$.

Proof. Each statement holds componentwise by Lemma 2.2, and the quotient map is compatible with all of them. □

2.1.1 *NAK* decompositions

Lemma 2.4. *$\overline{G(\mathbb{Q}_p)} = N(\mathbb{Q}_p) \cdot \overline{A(\mathbb{Q}_p)} \cdot \overline{K_p}$, with $N(\mathbb{Q}_p)$ embedded by $n \mapsto (n,1)$. If $(n,1)\,\tilde{a}\,\tilde{k} = (n',1)\,\tilde{a}'\,\tilde{k}'$ are two such decompositions, then $p(\tilde{a}') \in p(\tilde{a})\,(A(\mathbb{Q}_p) \cap K_p)$. In particular $|t|_p := \prod_{v|p} |t_v|_v$ is well defined for $p(\tilde{a}) = a(t)$, and depends only on (g,ζ). For p odd one may take $\tilde{k} \in K_p^*$.*

Proof. The Iwasawa decomposition $G(\mathbb{Q}_p) = N(\mathbb{Q}_p)A(\mathbb{Q}_p)K_p$ holds, with a determined modulo $A(\mathbb{Q}_p) \cap K_p$, which is $\prod_{v|p} A(\mathcal{O}_v^\times)$ for $p < \infty$ and a product of copies of $\{\pm 1\}$ and of $\{\mathrm{diag}(z, \bar{z}) : |z| = 1\}$ for $p = \infty$. Since μ_2 is central and lies in $\overline{A(\mathbb{Q}_p)}$, this lifts to $\overline{G(\mathbb{Q}_p)}$. For p odd, $\overline{K_p} = K_p^* \mu_2$ and the μ_2 factor can be moved into $\tilde{a}$. The absolute value $|t|_p$ is trivial on $A(\mathbb{Q}_p) \cap K_p$. □

2.1.2 Heights

For $g = (g_v) \in G(\mathbb{Q}_p)$ put $\|g\|_p = \prod_{v|p} \|g_v\|_v$, where $\|g_v\|_v = \max(|a|_v, |b|_v, |c|_v, |d|_v)$ for v finite and $\|g_v\|_v = (|a|^2 + |b|^2 + |c|^2 + |d|^2)^{1/2}$, raised to the power $[k_v : \mathbb{R}]$, for v archimedean (here $|\cdot|$ is the usual absolute value). Extend to the cover by $\|(g,\zeta)\|_p = \|g\|_p$.

Define $H = H_B \colon \overline{G(\mathbb{Q}_p)} \to \mathbb{R}$ by

$$H(g,\zeta) = \log |t|_p = \sum_{v|p} \log |t_v|_v, \tag{2.5}$$

where $(g,\zeta) = (n,1)\,\tilde{a}\,\tilde{k}$ as in Lemma 2.4 and $p(\tilde{a}) = a(t)$. This is the product of the normalized absolute values over $v \mid p$, that is $|N_{k \otimes \mathbb{Q}_p / \mathbb{Q}_p}(t)|_p$, and not the maximum used in the 1995 notes; see Section 2.1.3.

Lemma 2.5. *(a) $H(\tilde{a}\tilde{a}') = H(\tilde{a}) + H(\tilde{a}')$ for $\tilde{a}, \tilde{a}' \in \overline{A(\mathbb{Q}_p)}$.*

(b) $H(\tilde{n}\, x\, \tilde{k}) = H(x)$ for $\tilde{n} \in \overline{N(\mathbb{Q}_p)}$, $x \in \overline{G(\mathbb{Q}_p)}$ and $\tilde{k} \in \overline{K_p}$.

(c) $|H(x)| \ll 1 + \log \|x\|_p$ for $x \in \overline{G(\mathbb{Q}_p)}$ [A2].

The global analog, including invariance under $B(\mathbb{Q})$ on the left, is Lemma 2.8.

2.1.3 Characteristic functions of Weyl chambers

Let $P_0 = B$; the standard parabolic subgroups are B and G. For a standard $P = MN_P$ let A_P be the split component of the center of M over $\mathbb{Q}$. Thus $A_G = 1$, because the center $\{\pm 1\}$ of $SL(2)$ is finite, and $A_B \cong \mathbb{G}_m$ is the image of $t \mapsto a(t)$, $t \in \mathbb{Q}^\times \subset k^\times$: the $\mathbb{Q}$-split part of the torus $A \cong \mathrm{Res}_{k/\mathbb{Q}}\,\mathbb{G}_m$, which has $\mathbb{Q}$-rank one.

The $\mathbb{Q}$-rational characters of A form $X(A)_\mathbb{Q} \cong \mathbb{Z}$, generated by $\chi_1 = N_{k/\mathbb{Q}} \circ t$. Put $\mathfrak{a} = \mathfrak{a}_B = \mathrm{Hom}(X(A)_\mathbb{Q}, \mathbb{R})$, identified with $\mathbb{R}$ by $X \mapsto X(\chi_1)$. Then H of (2.5) and its global analog are $\langle H(x), \chi_1\rangle = \log|\chi_1(\tilde{a})|$, which is why H uses the norm. The simple root is $\alpha(a(t)) = t^2$; restricted to A_B it is $\alpha = 2\chi_1$, so $\alpha(X) = 2X$ under $\mathfrak{a} \cong \mathbb{R}$. Let $\alpha^\vee = X_\alpha \in \mathfrak{a}$ be the coroot, so $X_\alpha = 1$ and $\alpha(\alpha^\vee) = 2$, and let $\varpi = \frac{1}{2}\alpha^\vee$ be the fundamental coweight, so $\alpha(\varpi) = 1$. (The 1995 notes call ϖ the dual root.)

Let $\mathfrak{a}^+ = \{X \in \mathfrak{a} \mid \alpha(X) > 0\}$ and $(\mathfrak{a}^*)^+ = \{\Lambda \in \mathfrak{a}^* \mid \Lambda(\varpi) > 0\}$, with characteristic functions τ_B and $\hat{\tau}_B$, and set $\tau_G = \hat{\tau}_G = 1$. For standard $P_1 \subset P_2$ let τ_1^2, $\hat{\tau}_1^2$ and σ_1^2 be as in [A2]. With $\mathfrak{a} \cong \mathbb{R}$:

$$\tau_B = \tau_B^G = \hat{\tau}_B = \hat{\tau}_B^G = \text{characteristic function of } \{u \in \mathbb{R} \mid u > 0\}, \qquad \tau_1^2 = \begin{cases} \tau_1, & P_2 = G, \\ 1, & P_2 \neq G, \end{cases} \tag{2.6}$$

$$\sigma_B^G = \tau_B, \qquad \sigma_B^B = 0, \qquad \sigma_G^G = 1. \tag{2.7}$$

2.1.4 Algebras of functions

We follow Li [Li I, Li IV]. A representation π of a cover is *genuine* if $\pi(1, \zeta) = \zeta$ for $\zeta \in \mu_2$, and a function f on the cover is *anti-genuine* if $f(g, \zeta) = \zeta^{-1} f(g, 1)$. Test functions are anti-genuine. Then $\pi(f) = \int f(\tilde{g})\pi(\tilde{g})\, d\tilde{g}$ picks up the factor $\int_{\mu_2} \zeta^{-1}\zeta\, d\zeta = 1$, so it is in general nonzero on genuine representations, and it vanishes on representations that are trivial on μ_2.

For $n = 2$, $\zeta^{-1} = \zeta$, so "anti-genuine" and "genuine" define the same functions, and the conventions of the 1995 notes (genuine functions in §2.1.4, $\pi(g, \zeta) = \zeta^{-1}\pi(g, 1)$ in §3.1) agree with Li's. We keep Li's terminology so that the formulas remain correct for general n; see Remark 2026.4.

Let $C_c^\infty(\overline{G(\mathbb{Q}_p)})_-$ be the space of anti-genuine f such that $g \mapsto f(g, 1)$ lies in $C_c^\infty(G(\mathbb{Q}_p))$. Let p be odd. The *spherical Hecke algebra* $\mathcal{H}(\overline{G(\mathbb{Q}_p)}, K_p^*)$ consists of the $f \in C_c^\infty(\overline{G(\mathbb{Q}_p)})_-$ with $f(\tilde{k}\, x\, \tilde{k}') = f(x)$ for $\tilde{k}, \tilde{k}' \in K_p^*$. Its unit element is

$$e_p(\tilde{k}\zeta) = \zeta^{-1} \ (\tilde{k} \in K_p^*,\ \zeta \in \mu_2), \qquad e_p = 0 \text{ off } \overline{K_p}. \tag{2.8}$$

The function $(k, \zeta) \mapsto \zeta^{-1}$ on $\overline{K_p}$, which is what the identification $k \mapsto (k, 1)$ of the 1995 notes suggests, is $\prod_v \kappa_v \cdot e_p$; it is not bi-$K_p^*$-invariant.

2.1.5 Measures

Let dk be the Haar measure on K_p of total volume 1. Fix a nontrivial additive character ψ_v of k_v for each v, and let dn be the measure on $N(\mathbb{Q}_p) \cong \bigoplus_{v|p} k_v$ that is the product of the self-dual measures for the ψ_v. Let da be a Haar measure on $A(\mathbb{Q}_p)$. If T is a maximal torus of G over $\mathbb{Q}_p$, and $h \in G(\overline{\mathbb{Q}}_p)$ conjugates T to A, let dt be the pull-back of da as in [R]; stably conjugate tori get compatible measures as in [R]. Put

$$\delta_B(a(t)) = |\alpha(a(t))|_p = |t|_p^2.$$

Normalize the Haar measure dg on $G(\mathbb{Q}_p)$ by

$$\int_{G(\mathbb{Q}_p)} \phi(g)\,dg = \int_N \int_A \int_K \phi(nak)\,\delta_B(a)^{-1}\,dk\,da\,dn = \int_K \int_A \int_N \phi(kan)\,\delta_B(a)\,dn\,da\,dk. \tag{2.9}$$

The two expressions agree because dg is invariant under $g \mapsto g^{-1}$ (as G is unimodular) and $(nak)^{-1} = k^{-1}a^{-1}n^{-1}$; see [Kn, Ch. VIII] for the real case. (The 1995 notes have $\delta_B(a)^{1/2}$ for $g = kan$; the exponent must be 1 for $g = kan$ and -1 for $g = nak$.)

On a cover we use $d\tilde{g} = dg\,d\zeta$, where $d\zeta$ gives μ_2 total volume 1. Then $\overline{K_p}$ has volume 1, and e_p of (2.8) is idempotent: for $x = \tilde{k} \in K_p^*$,

$$(e_p * e_p)(x) = \int_{\overline{K_p}} e_p(y)\,e_p(y^{-1}x)\,dy = \int_{K_p^*} \int_{\mu_2} \zeta^{-1}\zeta\,d\zeta\,dy = 1.$$

2.2 Global aspects

The adelic cover is not the cover defined by $\prod_v \beta_v$: that product does not converge, because β_v is not trivial on $K_v \times K_v$ (Lemma 2.2(b)). The standard remedy [K2, Ge2, KP], used here, is to correct β_v by the coboundary of κ_v. This choice also fixes the splitting over $G(\mathbb{Q})$.

For each place v of k extend κ_v to $SL(2,k_v)$ by

$$\kappa_v(g) = \begin{cases} (c,d)_v, & v \text{ finite non-dyadic},\ g \in K_v,\ 0 < |c|_v < 1, \\ 1, & \text{otherwise}, \end{cases}$$

so $\kappa_v \equiv 1$ at dyadic and archimedean places, and put

$$\sigma_v(g_1,g_2) = \beta_v(g_1,g_2)\,\kappa_v(g_1)\kappa_v(g_2)\kappa_v(g_1g_2). \tag{2.10}$$

This cocycle is cohomologous to β_v; it defines the same cover $\overline{SL(2,k_v)}$, identified with the β_v-model by $(g,\zeta)_\sigma \mapsto (g, \kappa_v(g)\zeta)_\beta$. By Lemma 2.2(b), $\sigma_v \equiv 1$ on $K_v \times K_v$ at every finite non-dyadic v. Since $g \in G(\mathbb{A})$ has $g_v \in K_v$ for almost all v, the product

$$\sigma_{\mathbb{A}}(g_1,g_2) = \prod_v \sigma_v(g_{1,v}, g_{2,v}) \tag{2.11}$$

has only finitely many factors different from 1. It is a continuous 2-cocycle on $G(\mathbb{A}) = SL(2,\mathbb{A}_k)$ and defines

$$1 \to \mu_2 \to \overline{G(\mathbb{A})} \xrightarrow{p} G(\mathbb{A}) \to 1.$$

Equivalently, $\overline{G(\mathbb{A})}$ is the restricted product of the $\overline{SL(2,k_v)}$ with respect to the subgroups K_v^*, divided by $\{(\zeta_v) \in \bigoplus_v \mu_2 \mid \prod_v \zeta_v = 1\}$. It is a *quotient* of the restricted product, not a subgroup. Factorization of genuine representations of $\overline{G(\mathbb{A})}$ (and of $\overline{A(\mathbb{A})}$ in Section 3) refers to this quotient. Group law, inverses and conjugation are given by (2.2) and (2.3) with $\sigma_{\mathbb{A}}$ in place of β. For $H \subset G(\mathbb{A})$ put $\overline{H} = p^{-1}(H)$.

For $\gamma \in SL(2,k)$ we have $\gamma \in K_v$ and $\mathrm{ord}_v(c) \le 0$ or $c = 0$ for almost all v, so

$$\kappa_{\mathbb{A}}(\gamma) = \prod_v \kappa_v(\gamma) = \prod_{\substack{v \text{ finite non-dyadic} \\ \gamma \in K_v,\ 0<|c|_v<1}} (c,d)_v \tag{2.12}$$

is a finite product.

Lemma 2.6. *(a) The map $\gamma \mapsto (\gamma, \kappa_{\mathbb{A}}(\gamma))$ is a homomorphism $G(\mathbb{Q}) \to \overline{G(\mathbb{A})}$, so $\overline{G(\mathbb{Q})} = G(\mathbb{Q}) \times \mu_2$.*

(b) $n \mapsto (n, 1)$ splits $\overline{N(\mathbb{A})}$, and on $B(\mathbb{Q})$ the splitting of (a) is $b \mapsto (b, 1)$.

(c) Let S be a finite set of places containing the archimedean and dyadic places, and let $K_f = \prod_{v \in S \smallsetminus \Sigma_k^\infty} K_{m_v,v} \times \prod_{v \notin S} K_v$, with $m_v \geq 1$, and $m_v \geq 2e_v + 1$ if v is dyadic. Then $k \mapsto (k, 1)$ is a homomorphism $K_f \to \overline{G(\mathbb{A})}$; we write K_f^* for its image. (At $v \notin S$, translated to the β_v-model, its component is K_v^*.)

(d) The center of $\overline{G(\mathbb{A})}$ is $\overline{Z(G(\mathbb{A}))}$, where $Z(G(\mathbb{A})) = \prod_v \{\pm 1\}$. It is abelian, and a lift of $\varepsilon = (\varepsilon_v)$ has square $\prod_{v:\, \varepsilon_v = -1} (-1, -1)_v$, a finite product. It contains the subgroup $Z(G(\mathbb{Q})) \times \mu_2$ of order 4, a direct product, embedded by (a). It is not the direct product $Z(G(\mathbb{A})) \times \mu_2$ when k has a real place.

(e) $\sigma_{\mathbb{A}}(a(t_1), a(t_2)) = \prod_v (t_{1,v}, t_{2,v})_v$ for $t_1, t_2 \in \mathbb{A}_k^\times$, and the preimage of every abelian subgroup of $G(\mathbb{A})$, in particular $\overline{A(\mathbb{A})}$, is abelian.

Proof. (a) For $\gamma_1, \gamma_2 \in SL(2, k)$, $\prod_v \beta_v(\gamma_1, \gamma_2) = 1$ by the product formula (H4), because $\beta_v(\gamma_1, \gamma_2)$ is the local Hilbert symbol of two elements of $k^\times$. Hence $\sigma_{\mathbb{A}}(\gamma_1, \gamma_2) = \kappa_{\mathbb{A}}(\gamma_1)\kappa_{\mathbb{A}}(\gamma_2)\kappa_{\mathbb{A}}(\gamma_1\gamma_2)$, which is the condition for $\gamma \mapsto (\gamma, \kappa_{\mathbb{A}}(\gamma))$ to be a homomorphism.

(b) On N every β_v and κ_v is 1. On $B(\mathbb{Q})$ every κ_v is 1 because $c = 0$.

(c) It suffices that $\sigma_v \equiv 1$ on the v-component of $K_f \times K_f$ for each finite v. At $v \notin S$ this is Lemma 2.2(b). At finite $v \in S$, $\kappa_v \equiv 1$ on $K_{m_v,v}$ (non-dyadic: $c \in \varpi\mathcal{O}$ and d is a square; dyadic: by definition), so $\sigma_v = \beta_v \equiv 1$ there by Lemma 2.2(c).

(d) Lemma 2.2(g) applies to σ_v, since $\kappa_v(g_1)\kappa_v(g_2)\kappa_v(g_1g_2)$ is symmetric in commuting g_1, g_2; so preimages of abelian subgroups are abelian, and $\overline{Z(G(\mathbb{A}))}$ is central. Its image under p is central, so equality holds. Since $\kappa_v(\pm 1) = 1$, the square of a lift of ε is $\prod_v \sigma_v(\varepsilon_v, \varepsilon_v) = \prod_{\varepsilon_v = -1} (-1, -1)_v$, which is finite by (H3). For $\varepsilon = -1 \in Z(G(\mathbb{Q}))$ it is $\prod_v (-1, -1)_v = 1$ by (H4). If v_0 is real, the lift of the element that is -1 at v_0 and 1 elsewhere has order 4.

(e) As in Lemma 2.2(f), since $\kappa_v \equiv 1$ on A. □

From now on we identify $G(\mathbb{Q})$ with its image under Lemma 2.6(a), and $N(\mathbb{A})$ with $N(\mathbb{A}) \times 1$. By (b), the two identifications agree on $N(\mathbb{Q})$. The splitting over $B(\mathbb{Q})$ by $b \mapsto (b, 1)$ is what the constant terms, Eisenstein series and the truncation operator of Sections 4 and 5 use.

Example 2.B. Let $k = \mathbb{Q}$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$ with $c \neq 0$. Then $\gamma \in K_v$ for every v, and for odd $p \mid c$, $(c, d)_p = \left(\frac{d}{p}\right)^{\mathrm{ord}_p(c)}$ by (H3). So

$$\kappa_{\mathbb{A}}(\gamma) = \left(\frac{d}{c'}\right), \qquad c' = \text{the odd part of } |c|,$$

a Jacobi symbol: the factor behind the theta multiplier on $\Gamma_0(4)$ [K2, Ge2]. The script checks this on random elements of $SL(2, \mathbb{Z})$.

Remark 2026.2. The 1995 notes define the splitting as $\gamma \mapsto (\gamma, s_{\mathbb{A}}(\gamma)^{-1})$, with $s_v(\gamma)^{-1} = (c, d)_v$ when v is finite, $cd \neq 0$ and $n \nmid \mathrm{ord}_v(c)$, and with the cocycle $\prod_v \beta_v$. That cocycle does not converge, and the formula belongs to a different normalization. In the normalization (2.11), $\kappa_{\mathbb{A}}$ of (2.12) replaces $s_{\mathbb{A}}^{-1}$. For $n = 2$ the two agree at every non-dyadic v with $\gamma \in K_v$: if $\mathrm{ord}_v(c) > 0$

is even, or if $\mathrm{ord}_v(c) = 0$, both $(c,d)_v$ and $\kappa_v(\gamma)$ are 1. They differ only at the finitely many places where $\gamma \notin K_v$, and at dyadic places.

Let $K_0 = \prod_v K_v$ be the standard maximal compact subgroup of $G(\mathbb{A})$, with K_v as in Section 2.1; K_f and K_∞ denote its finite and archimedean parts.

Lemma 2.7 (Strong approximation; compare [K2, Prop. 5], [Ge2])**.** *Let K_f and K_f^* be as in Lemma 2.6(c), and put $\Gamma = G(\mathbb{Q}) \cap K_f$, a congruence subgroup of $SL(2,\mathcal{O}_k)$, viewed in $G(\mathbb{R}) = SL(2, k\otimes\mathbb{R})$. For $\gamma \in \Gamma$ let $\tilde{k}_\gamma \in K_f^*$ be the lift of γ_f. Then $\tilde{\gamma} = \gamma\,\tilde{k}_\gamma^{-1}$ lies in $\overline{G(\mathbb{R})}$, the map $\gamma \mapsto \tilde{\gamma}$ is an injective homomorphism, its image $\widetilde{\Gamma}$ is discrete of finite covolume, and*

$$G(\mathbb{Q})\backslash\overline{G(\mathbb{A})}/K_f^* \;\cong\; \widetilde{\Gamma}\backslash\overline{G(\mathbb{R})}.$$

Proof. Since $SL(2)$ is simply connected and $G(\mathbb{R})$ is not compact, strong approximation gives $G(\mathbb{A}) = G(\mathbb{Q})\,G(\mathbb{R})\,K_f$. Lifting to the cover, with $\mu_2 \subset \overline{G(\mathbb{R})}$, gives $\overline{G(\mathbb{A})} = G(\mathbb{Q})\,\overline{G(\mathbb{R})}\,K_f^*$, so the double coset space has a single component. The stabilizer computation is the usual one: $\gamma \in G(\mathbb{Q})$ lies in $\overline{G(\mathbb{R})}K_f^*$ exactly when $\gamma_f \in K_f$, and then $\gamma = \tilde{\gamma}\,\tilde{k}_\gamma$. The map $\gamma \mapsto \tilde{\gamma}$ is a homomorphism because lifts of elements of $G(\mathbb{A})$ with disjoint supports commute: $\sigma_v(1,h) = \sigma_v(h,1) = 1$. □

2.2.1 *NAK* decompositions

Lemma 2.8. *(a) $\overline{G(\mathbb{A})} = N(\mathbb{A})\cdot\overline{A(\mathbb{A})}\cdot\overline{K_0}$. In a decomposition $(n,1)\,\tilde{a}\,\tilde{k}$, $p(\tilde{a})$ is determined modulo $A(\mathbb{A})\cap K_0$, and*

$$H(g,\zeta) = \log|t|_{\mathbb{A}_k} = \sum_v \log|t_v|_v, \qquad p(\tilde{a}) = a(t),$$

is well defined.

(b) H satisfies Lemma 2.5(a)–(c) globally, and $H(\gamma x) = H(x)$ for $\gamma \in B(\mathbb{Q})$ and $x \in \overline{G(\mathbb{A})}$.

Proof. (a) As for Lemma 2.4, place by place. (b) For $\gamma = n'a(s) \in B(\mathbb{Q})$, identified with $(\gamma,1)$ by Lemma 2.6(b), and $x = (n,1)\tilde{a}\tilde{k}$, $\gamma x = (n'',1)\,(a(s),1)\tilde{a}\,\tilde{k}$ with $n'' \in N(\mathbb{A})$, so $H(\gamma x) = \log|s|_{\mathbb{A}_k} + H(x) = H(x)$ by the product formula. □

2.2.2 Heights

For $g = (g_v) \in G(\mathbb{A})$ put $\|g\| = \prod_v \|g_v\|_v$ and $\|(g,\zeta)\| = \|g\|$.

Lemma 2.9. *For $x, y \in \overline{G(\mathbb{A})}$ and $\tilde{k}, \tilde{k}' \in \overline{K_0}$: (a) $\|x\| \geq 1$; (b) $\|\tilde{k}\,x\,\tilde{k}'\| = \|x\|$; (c) $\|xy\| \leq \|x\|\,\|y\|$; (d) $\|x^{-1}\| = \|x\|$.*

Proof. Each holds for $G(\mathbb{A})$ place by place, since $\det g_v = 1$ forces $\|g_v\|_v \geq 1$. For (d), g^{-1} has the same entries as g up to order and sign. The cover does not change $\|\cdot\|$. □

2.2.3 Algebras of functions

A function on $\overline{G(\mathbb{A})}$ is *anti-genuine* if $f(g,\zeta) = \zeta^{-1}f(g,1)$. Let $\mathcal{H}(\overline{G(\mathbb{A})})$ be the span of the functions $f = \bigotimes_v f_v$, $f_v \in C_c^\infty(\overline{SL(2,k_v)})_-$, with f_v equal to the unit element e_v of (2.8) (defined with K_v^*) for almost all v. Such a product is a function on the restricted product of the local covers which is invariant under $\{(\zeta_v) \mid \prod\zeta_v = 1\}$, because each f_v is anti-genuine; so it descends to $\overline{G(\mathbb{A})}$ and is anti-genuine there.

2.2.4 Measures

Let $dk = \prod_v dk_v$ on K_0. Let $dg = \prod_v dg_v$ be the Tamagawa measure on $G(\mathbb{A})$ [W2], with dg_v built from the self-dual measures of a fixed nontrivial character $\psi = \prod \psi_v$ of $\mathbb{A}_k/k$; then $\mathrm{vol}(G(\mathbb{Q})\backslash G(\mathbb{A})) = 1$. For a maximal torus T of G over $\mathbb{Q}$ let dt be the Tamagawa measure on $T(\mathbb{A})$, written da for $T = A$, and let dn be the product measure on $N(\mathbb{A})$, for which $\mathrm{vol}(N(\mathbb{Q})\backslash N(\mathbb{A})) = 1$. On the cover put $d\tilde{g} = dg\, d\zeta$ with $\mathrm{vol}(\mu_2) = 1$, so $\mathrm{vol}(G(\mathbb{Q})\backslash \overline{G(\mathbb{A})}) = 1$.

2.3 The double cover of $GL(2)$

Section 7 compares $SL(2)$-conjugacy with $GL(2)$-conjugacy on the cover, so we need a double cover of $GL(2)$ which restricts to the one above. There are two natural choices for $n = 2$ (Remark 2026.3); we fix one, and compute the conjugation action of $GL(2,F)$ on $\overline{SL(2,F)}$ for later use.

For $g_1, g_2 \in GL(2,F)$, with $x(g)$ defined as before, put

$$\sigma(g_1,g_2) = \Big(\frac{x(g_1g_2)}{x(g_1)},\ \frac{x(g_1g_2)}{x(g_2)\det g_1}\Big)_v, \tag{2.13}$$

the $n = 2$ case of Kubota's cocycle on $GL(2)$ [K2, Ge2]. Let $\overline{GL(2,F)}$ be the corresponding extension by μ_2.

Lemma 2.10. *(a) σ is a 2-cocycle on $GL(2,F)$ whose restriction to $SL(2,F)$ is β. Hence $p^{-1}(SL(2,F)) \subset \overline{GL(2,F)}$ is the cover $\overline{SL(2,F)}$ of Section 2.1.*

(b) For v finite non-dyadic, $k \mapsto (k, \kappa'(k))$ is a homomorphism on $GL(2,\mathcal{O})$, where $\kappa'(k) = (c,\, d/\det k)_v$ if $0 < |c|_v < 1$ and $\kappa'(k) = 1$ otherwise. On $SL(2,\mathcal{O})$, $\kappa' = \kappa$.

(c) Conjugation by any lift of $h \in GL(2,F)$ defines an automorphism $\mathrm{Ad}(h)$ of $\overline{SL(2,F)}$ lifting conjugation by h, and $h \mapsto \mathrm{Ad}(h)$ is an action of $GL(2,F)$. If h commutes with $\gamma \in SL(2,F)$, then

$$\mathrm{Ad}(h)(\gamma,\zeta) = \Big(\gamma,\ \Big(\frac{x(h\gamma)}{x(h)},\ \det h\Big)_v \zeta\Big). \tag{2.14}$$

In particular $\mathrm{Ad}(\mathrm{diag}(y,1))(a(t),\zeta) = (a(t),(y,t)_v\zeta)$, so the preimage of the diagonal torus of $GL(2,F)$ is not abelian unless $F = \mathbb{C}$. Likewise $E^\times \supset T_\theta$ (embedded as $\{\begin{pmatrix} a & b\theta \\ b & a \end{pmatrix}\}$) acts on $\overline{T_\theta}$ through (2.14).

Proof. (a) On $SL(2)$, $\det g_1 = 1$ and (2.13) becomes (2.1). The cocycle identity is due to Kubota [K2]; we have also checked it numerically.

(b) We have checked this numerically for $F = \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7$; see Remark 2026.5.

(c) Since μ_2 is central, conjugation by (h,ζ) does not depend on ζ, and it preserves $p^{-1}(SL(2,F))$. If $h\gamma = \gamma h$, then $\mathrm{Ad}(h)(\gamma,\zeta) = (\gamma, \zeta\,\sigma(h,\gamma)\sigma(\gamma,h))$. With $x_3 = x(h\gamma)$, bimultiplicativity gives $\sigma(h,\gamma) = (x_3/x(h), x_3/x(\gamma))\,(x_3/x(h), \det h)$, and $\sigma(\gamma,h) = (x_3/x(\gamma), x_3/x(h))$ since $\det\gamma = 1$. By symmetry (H1) the first factors cancel. For $h = \mathrm{diag}(y,1)$ and $\gamma = a(t)$: $x(h\gamma) = t^{-1}$ and $x(h) = 1$, giving $(t^{-1},y) = (y,t)$. This is nontrivial for suitable y,t unless $F = \mathbb{C}$. □

The consequences for Section 7 (the functions $\delta \mapsto I_{\mathcal{O}}(f^\delta)$ are defined on $D(T/F)$ only up to the character (2.14), and the choice of $D(A/F)$) are taken up there.

Remark 2026.3. The cocycle $\sigma'(g_1,g_2) = \sigma(g_1,g_2)\,(\det g_1, \det g_2)_v$ is another double cover of $GL(2,F)$ with the same restriction to $SL(2,F)$; these are the two Kazhdan–Patterson covers for $n=2$ [KP]. We fix σ. Whether Section 7 needs σ' instead has not been checked at this stage. The commutator formula (2.14) is the same for σ': the extra factor $(\det h, \det\gamma)(\det\gamma, \det h)$ is 1 by (H1). The adelic cover of $GL(2,\mathbb{A}_k)$ is built as in Section 2.2, with κ'_v in place of κ_v; Hilbert reciprocity for $\prod_v \sigma_v$ on $GL(2,k)$ is checked numerically for $k=\mathbb{Q}$.

Remark 2026.4. For general n (with $\mu_n \subset k$) the following change. The conjugation action of $\overline{A(F)}$ on itself is through $(t_1,t_2)_n^2$, so $\overline{A(F)}$ is abelian only for $n \le 2$; the argument of Lemma 2.2(g) uses the symmetry of the quadratic symbol and does not extend. Lemma 2.2(c) holds with $1+\varpi^m\mathcal{O} \subset F^{\times n}$ (the same proof). Genuine (ζ) and anti-genuine (ζ^{-1}) differ, and with the 1995 convention $R(f)$ vanishes on L^2_{gen} unless $n \mid 2$.

Remark 2026.5. The script `checks/sec2_n2_check.py` verifies, in exact rational arithmetic at $v = \infty, 2, 3, 5, 7$: the cocycle identity and the two forms of (2.1); Lemma 2.2(a)–(c), (e)–(h); Remark 2026.1; the product formula and the splitting of Lemma 2.6(a), (b), (d); Example 2.B; and Lemma 2.10(a)–(c) with Remark 2026.3. The checks are on random elements of $SL(2,\mathbb{Q})$, $GL(2,\mathbb{Q})$, $SL(2,\mathbb{Z}_{(p)})$ and the tori $T_\theta(\mathbb{Q})$; they are evidence, not proofs. Of the statements in this section, Lemma 2.2(b) and Lemma 2.10(a), (b) are supported here only by citation and these checks.

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